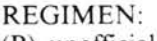
SECTION 1 Time—30 minutes 45 Questions

For each question in this section, choose the best answer and fill in the corresponding oval on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly



1. OBEDIENT: (A) bitter (B) unhappy (C) grotesque (D) unruly (E) dangerous

2. DIFFUSE: (A) bring together (B) make habitual (C) leave unattended (D) openly admire (E) seldom heed

3. PAMPER: (A) magnify (B) conceal (C) discipline (D) reclaim (E) confuse

4. PUCKERED: (A) ancient (B) smooth (C) zigzag (D) pale (E) thin

5. HYPOCRISY: (A) sincerity (B) remorse (C) enthusiasm (D) persistence (E) anxiety

6. GHASTLY: (A) appealing (B) noisy (C) generous (D) casual (E) approving

7. ANTIPATHY: (A) urgency (B) idiosyncrasy (C) competence (D) constant repetition (E) strong affection

8. PERTURBATION: (A) hardiness (B) alacrity (C) frankness (D) serenity (E) obstinacy

9. CONSTRICT: (A) suspend (B) regard (C) invert (D) distend (E) exclude

10. SANCTIFY: (A) amplify (B) threaten (C) taint (D) separate (E) release

11. REGIMEN: (A) unpopular policy (B) unofficial agent (C) impractical solution (D) unsystematic activity (E) disaffected group

12. SOMBER: (A) light in color (B) quick to understand (C) unable to sleep (D) resistant to change (E) small in size

13. DECADENT: (A) enormously wealthy (B) remarkably charming (C) ruthless (D) distinctive (E) flourishing

14. INTRINSIC: (A) extensive (B) shoddy (C) divided (D) simple (E) extraneous

15. FECUND: (A) gradual (B) adaptable (C) external (D) untroubled (E) barren

GO ON TO THE NEXT PAGE

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Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that, when inserted in the sentence, best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. The strong affinity of these wild sheep for mountains is not ----: mountain slopes represent ---- because they effectively limit the ability of less agile predators to pursue the sheep.

(A) useful. .peril
(B) accidental. .security
(C) instinctive. .attainment
(D) restrained. .nourishment
(E) surprising. .inferiority

17. An author whose only ---- is a biography, he has been dismissed by its critics as a mere reporter who recorded the brilliant wit of another and thereby gained a ---- fame of his own.

(A) failure. .belated
(B) justification. .tragic
(C) accomplishment. .spurious
(D) collaboration. .complete
(E) enterprise. .deserved

18. Hastings is deeply contemptuous of the ---- world; his notions have obliged him to live a life of ----.

(A) practical. .utility (B) sensual. .gratification (C) ethereal. .inspiration (D) public. .solitude (E) depraved. .abandon

19. Convinced that they themselves ---- relevant knowledge, most of that country's citizens consigned political judgments to experts and technocrats, thereby reducing themselves to ----.

(A) commanded. .tyrants
(B) possessed. .witnesses
(C) disseminated. .automatons
(D) required. .representatives
(E) lacked. .spectators

20. In some animal species, differences between opposite sexes are so ---- that it is very difficult to tell that the male and female are ----.

(A) measurable. .distinct
(B) minute. .similar
(C) obvious. .indistinguishable
(D) extreme. .related
(E) trivial. .identical

21. In her opening statement, the defense attorney told the jury that she would prove how one fact that appears to conflict with another may actually be ---- it.

(A) superior to (B) contradictory to (C) detrimental to (D) incompatible with (E) reconcilable with

22. If his methods were autocratic and his manner imperious, these were ---- and frequently even ---- as attributes becoming a monarch.

(A) criticized. .glorified
(B) overlooked. .ignored
(C) discussed. .denounced
(D) tolerated. .applauded
(E) encouraged. .dismissed

23. A fear of ----, prevalent among nineteenth-century educators, was evident in their frequent exhortations on the need to protect adolescents from the premature acquisition of knowledge and abilities.

(A) isolation (B) illiteracy (C) precocity (D) barbarity (E) frivolity

24. Because Alvin Ailey's choreography combines elements from traditions as different from each other as African folk dance and classical European ballet, his approach has been aptly termed ----.

(A) eclectic (B) intense (C) impromptu (D) theoretical (E) conventional

25. Like most political ---- in the United States, this newly coined phrase designates a ---- of groups and programs whose parts do not fit together seamlessly.

(A) labels. .mosaic (B) symbols. .symmetry (C) parties. .discord (D) slogans. .milestone (E) entities. .violation

GO ON TO THE NEXT PAGE

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

Two kinds of electric force operate between atoms or molecules: an attractive force, which is effective over a limited range (a few times the atom's or molecule's diameter), and a highly repulsive force, which is effective over a much shorter range. In the solid state, the forces between atoms or molecules determine the structure. When two atoms or molecules are about one hundred-millionth of an inch apart, these two forces cancel each other out, establishing a point of equilibrium and causing atoms or molecules to vibrate slightly toward and away from each other, as if held by invisible springs.

A small bar of iron looks solid, but the component atoms of the iron are entirely separate, suspended in space. Each of the individual atoms is vibrating unceasingly toward and away from each one of its neighbors. As they hang in space, the atoms form the intersections of a regular latticework. In iron, the lattice is composed of tiny cubes with an atom at each corner and one in the middle. Other substances have lattices with sides forming a variety of geometrical shapes which give each element its peculiar internal structure. When an object is broken, it normally separates along the lines of the lattice.

An increase in temperature speeds up the average velocity of molecular vibration within solids just as it does in gases. Each swinging movement of a single molecule, under the influence of added heat, is wider and faster and increases the average molecular separation. In other words, a solid object generally expands as a whole when the temperature is increased. As the additional heat forces the vibrating movement of the tightly bound molecules to become wilder and wilder, a point is reached at which some of the molecules begin to burst away from the binding forces exerted by their neighbors—they slip out of the regular latticework. With further heat, more molecules escape their captivity. As the rigid order begins to diminish, whole clusters of molecules begin to slip past each other. Finally the substance is no longer a solid, and the resultant breakdown of order is a process called melting.

When internal fluidity is fully achieved, the molecules are said to be in the liquid state. In liquids, the molecules hurtle in and then out of each other's areas of influence. Thus the force of a single molecule on another can be exerted for only the briefest time. Neighbors appear and disappear in millionths of a second. In liquids, as in solids, the addition of heat increases the average space between molecular centers. The observation that both liquids and solids generally expand with the addition of heat is best explained by the kinetic theory.

The reading passages in this test are brief excerpts or adaptations of excerpts from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

26. The primary focus of the passage is on

(A) how liquid substances alter when changing to the solid state
(B) how forces among molecules determine the structure of matter
(C) the shapes disrupted by lattices of atoms
(D) the advantages of equilibrium in liquids
(E) the dominance of the repulsive force in the solid state

27. Which of the following best describes what happens to the latticework of iron when iron changes from a solid to a liquid state?

(A) It recombines into larger cubes.
(B) Its sides shift to form other geometric shapes.
(C) It vibrates.
(D) It shrinks.
(E) It disappears.

28. According to the passage, a substance is very near its melting point when which of the following occurs?

(A) A change in temperature speeds up the average velocity of molecular vibration.
(B) Whole clusters of molecules begin to slip past one another.
(C) An object separates along the lines of the lattice.
(D) The attractive and repulsive forces cancel each other out.
(E) The component atoms are completely separated from one another.

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29. It can be inferred from the passage that internal fluidity has been fully achieved when which of the following conditions exists?

    (A) More and more molecules escape their captivity.
    (B) Heat increases the average space between molecular centers.
    (C) Forces are effectively exerted by any particular molecule on another for only a very brief time.
    (D) The vibrating movement of the tightly bound molecules becomes wilder and wilder.
    (E) The lattice assumes a different geometric shape.

30. The passage indicates that which of the following statements is true of both a solid and a liquid?

    (A) The addition of heat generally increases the average space between molecular centers.
    (B) The lattice is composed of tiny cubes with an atom at each corner and one in the middle.
    (C) The molecules hurtle in and then out of each other's areas of influence.
    (D) Each swinging movement of a single molecule is wider and wider.
    (E) The attractive force is always stronger than the repulsive force.

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The constant reference point for some Chicano historians is the Indian heritage, whether it be Mayan, Aztec, or Pueblo. Because Chicanos are mestizos, they possess an Indian background; I do not quarrel with this. What I do object to is the rejection of our European past. The history of America since the European conquest has been shaped by European factors adjusted to the American environment. Chicanos are a product of this adjustment, and we cannot understand our culture or our history without recognizing this fact.

This is not to suggest that we should go to the extreme of calling ourselves Spanish, Spanish Americans, or Hispanos. The fact is we are Mexicans. This distinction was overlooked by the historian Bolton, for example. In an article published in 1930, he made the point that the Southwest was the "meeting place and fusing place of two streams of European civilization, one coming from the south (the Spanish), the other from the north (the English)." In describing the Spanish influence there, Bolton mentioned the fact that states, rivers, mountains possessed Spanish names. "Many towns have Spanish quarters," he wrote, "where the life of the old days goes on, and where one can always hear the soft Castilian tongue."

Now, Chicanos know there are very few, if any, places in the Southwest where the "soft Castilian tongue" can be heard! The point, of course, is that Bolton did not recognize that the civilization existing in the Southwest at the time of the United States conquest in 1848 was not Spanish, but Mexican. It was, therefore, a mestizo society—one that had fused the Indian elements with the Spanish ones to create a distinct national culture. De Gandia recognizes the existence of national cultures, such as the Mexican, but at the same time acknowledges that not only the culture of Mexico, but also the political, economic, and social currents of the Mexican experience were for the most part European in origin. And this is what we have to understand.

We can be proud of being Mexican or Chicano and, at the same time, recognize and try to understand the European influence, thereby coming to know ourselves better. I am not saying we should be proud of our European heritage—that is up to each individual. What I am saying is that to neglect it is to neglect reality.

Leopoldo Zea, the Mexican philosopher, writes that "Mexicanism in itself cannot be a legitimate goal, but only a point of departure, a means toward a broader and more responsible task." What Zea says about "Mexicanism" is also true for "Chicanismo"; it is not a goal, but a means. And, I submit, it is a means toward a universal understanding of people. We look at ourselves and at others for the purpose of comprehending the human condition.

31. The passage primarily addresses itself to which of the following questions?

(A) What is the most appropriate cultural perspective for the study of Chicano history?
(B) Which cultural tradition of the Chicano heritage is the most interesting?
(C) How are Chicano writers influenced by their Indian heritage?
(D) Why should scholars learn to distinguish between "Mexicanism" and "Chicanismo"?
(E) Which of the Indian traditions most directly influenced Chicano history?

32. The author's position on the European heritage of Chicano culture can be described as which of the following?

(A) An assertion of its paramount importance to Indian culture
(B) A denial of its usefulness as an aid to understanding Chicano heritage
(C) A recognition of its relevance to Chicano history
(D) An analysis of the values and goals that it represents
(E) An acknowledgment of its pervasive influence and praise for its contributions

33. The passage suggests that the author would probably consider which of the following most helpful in presenting a realistic account of Chicano history?

(A) An essay that traced elements of Indian culture in modern Mexico
(B) A book that examined the interaction between Indian and European cultures in Mexico
(C) An article that celebrated the Indian civilization before the Colonial era
(D) A diary that recorded the life of a young Chicano a hundred years ago
(E) A dissertation that listed all known sites of ancient Mayan cities

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34. The author's discussion of Bolton develops the primary point of the passage by

(A) emphasizing the inaccuracy of a culturally narrow view of Chicano history
(B) demonstrating the direct relevance of European culture to Chicano identity
(C) illustrating the temptation of most historians to misrepresent reality
(D) questioning the importance of varying linguistic patterns
(E) affirming the necessity of discussion between European and Indian scholars

35. Which of the following best describes the function of the last paragraph in the passage?

(A) It adds further details to the statements made previously.
(B) It defines previously implicit assumptions about the European heritage.
(C) It presents a personal view of historical facts presented earlier.
(D) It illustrates the passage's generalizations about Chicano culture.
(E) It justifies the field of study discussed in the rest of the passage.

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN:BOREDOM :: (A) dream:sleep (B) anger:madness (C) smile:amusement (D) face:expression (E) impatience:rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. NEEDLE : KNITTING :: (A) finger : sewing (B) sign : painting (C) throat : singing (D) hurdle : running (E) chisel : carving

37. SUBMERGE : WATER :: (A) parch : soil (B) bury : earth (C) suffocate : air (D) disperse : gas (E) extinguish : fire

38. APPAREL : SHIRT :: (A) sheep : wool (B) foot : shoe (C) light : camera (D) belt : buckle (E) jewelry : ring

39. INFURIATE : DISPLEASE :: (A) release : drop (B) oppress : swelter (C) drench : moisten (D) stir : respond (E) conceive : imagine

40. PROPHETIC : FUTURE ::
(A) incredible : belief
(B) contemporary : time
(C) visionary : reality
(D) historical : past
(E) revolutionary : fashion

41. STRATAGEM : OUTWIT ::
(A) prototype : design
(B) variation : change
(C) decoy : lure
(D) riddle : solve
(E) charade : guess

42. SHACKLE : MOVEMENT :: (A) weapon : protection (B) anchor : ship (C) rudder : direction (D) alarm : warning (E) gag : speech

43. CHARLATAN : EXPERTISE :: (A) glutton : food (B) orator : lecture (C) braggart : impression (D) liar : truth (E) performer : idleness

44. WANDERLUST : TRAVEL :: (A) fantasy : indulge (B) innocence : confess (C) ignorance : know (D) digression : speak (E) avarice : acquire

45. EXTRICATE : QUAGMIRE :: (A) liberate : captivity (B) rescue : lifeboat (C) defer : decision (D) eradicate : swamp (E) strain : filter

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

SECTION 2 Time—30 minutes, 25 Questions

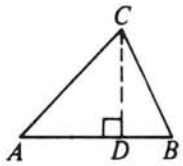
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and fill in the corresponding oval on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definition of symbols:

$=$ is equal to	$\leqq$ is less than or equal to
$\neq$ is unequal to	$\geqq$ is greater than or equal to
$<$ is less than	$\parallel$ is parallel to
$>$ is greater than	$\perp$ is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.
If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

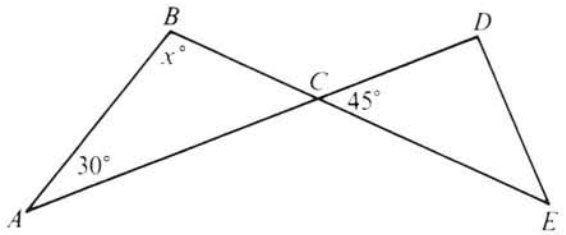
Note: Figures that accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

1. Bill weighs exactly twice as much as his brother Fred. If Bill's weight is between 95 and 120 pounds, then Fred's weight CANNOT be equal to which of the following?

(A) 47
(B) 50
(C) 52
(D) 55
(E) 57

2. Ten oranges cost \$1.25. At this rate, if 2 oranges are purchased and paid for with a \$5.00 bill, what is the correct change? (Assume no tax.)

(A) \$0.75
(B) \$1.25
(C) \$3.50
(D) \$3.75
(E) \$4.75



3. In the figure above, what is the value of x ?

(A) 75
(B) 90
(C) 95
(D) 105
(E) 110

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4. A man travels 150 miles in 3 hours. If he continues at this rate, how many miles will he travel in the next 2 hours?

(A) 50
(B) 100
(C) 250
(D) 300
(E) 900

5. $\frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{4}{5} \cdot \frac{5}{6} \cdot \frac{6}{7} =$

(A) $\frac{1}{7}$

(B) $\frac{3}{7}$

(C) $\frac{21}{27}$

(D) $\frac{6}{7}$

(E) $\frac{7}{8}$

6. If $x = 2$ and $y = -1$, what is the value of $\frac{x - y}{x + y}$?

(A) -3

(B) -1

(C) $\frac{1}{3}$

(D) 1

(E) 3

x	$2x - 1$
0	-1
2	3
a	11

7. In the table shown above, $a =$

(A) 21
(B) 20
(C) 10
(D) 6
(E) 4

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8. $\sqrt{25 - 9} =$

(A) 2
(B) 4
(C) 8
(D) 16
(E) 32

9. If $6{,}565 = 65(x + 1)$, then $x =$

(A) 10
(B) 11
(C) 100
(D) 101
(E) 1,001

10. A number is formed by arranging the digits of 1,985 in descending order and another number is formed by arranging these digits in ascending order. The difference between these two newly formed numbers is

(A) 8,372
(B) 8,362
(C) 8,272
(D) 8,262
(E) 3,906

11. A drugstore had a promotion in which \$0.50 was refunded to customers for each \$10.00 spent on merchandise. If a customer spent \$20.00 on merchandise during this promotion, the refund was equal to what percent of the amount spent?

(A) $1\frac{1}{4}\%$

(B) $2\frac{1}{2}\%$

(C) 5%

(D) 25%

(E) 50%

12. In $\triangle RST$, $\angle S$ is a right angle and $\angle R$ is 10° greater than $\angle T$. What is the measure of $\angle T$?

(A) 40°
(B) 45°
(C) 50°
(D) 55°
(E) 80°

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13. Abby cuts each of $n + 3$ apple pies into 5 pieces. In terms of n, what is the total number of pieces of pie?

(A) $\frac{n}{5} + \frac{3}{5}$

(B) $\frac{n}{5} + 3$

(C) $n + 15$

(D) $5n + 3$

(E) $5n + 15$

14. For a certain hot water heater, the increase in heating expenses is directly proportional to the increase in water-temperature setting. If heating expenses increase by \$24 when the water-temperature setting increases by 20 degrees Fahrenheit, by how much will heating expenses increase if the water-temperature setting is increased by 15 degrees Fahrenheit?

(A) \$16
(B) \$18
(C) \$19
(D) \$20
(E) \$21

15. If an integer is added to its square, the units' digit of this sum could be

(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

16. If a circle with a circumference of 20 has two overlapping arcs with lengths of 16 and 14, respectively, what is the LEAST possible total length of the overlap of the two arcs?

(A) 2
(B) 4
(C) 6
(D) 10
(E) 12

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Questions 17-19 refer to the following information.

A computer is programmed to read a positive integer n and carry out the following instructions repeatedly until a value of 1 is obtained.

If n is even, then the new value of n is $\frac{n}{2}$.

If n is odd, then the new value of n is $3n + 1$.

For example, if the first value of n is 14, the second value of n is 7 and the third value is 22. The process continues and the results are printed in the following form.

value #1 = 14
value #2 = 7
value #3 = 22
. .
. .
. .

The process stops when a value of 1 is obtained.

17. If the first value of n is 40, what is the fifth value of n ?

(A) 2
(B) 5
(C) 10
(D) 16
(E) 91

18. For which of the following first values of n is the instruction "the new value of n is $3n + 1$" never carried out?

(A) 100
(B) 80
(C) 64
(D) 48
(E) 25

19. When the first value of n is 27, the 112th value is 1. When the first value of n is 54, which value is 1 ?

(A) 56th
(B) 58th
(C) 111th
(D) 113th
(E) 224th

20. If a circle with center at the origin passes through the point (0, 5), through which of the following points does it also pass?

(A) $(2, \sqrt{21})$
(B) (4, 5)
(C) $(5, \sqrt{21})$
(D) (5, 5)
(E) (5, 12)

21. If 12 eggs cost x cents and 20 slices of bacon cost y cents, what is the cost, in cents, of 2 eggs and 4 slices of bacon?

(A) $\frac{xy}{30}$
(B) $\frac{x + 2y}{30}$
(C) $\frac{5x + 3y}{60}$
(D) $\frac{5x + 6y}{30}$
(E) $\frac{6x + 5y}{30}$

22. Each "word" in a special code is a three-digit number. If no word in the code can begin with a 1 or a 0, what is the greatest number of different words the code can contain?

(A) 28
(B) 512
(C) 800
(D) 900
(E) 1,000

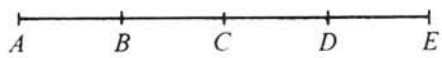
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23. In a windowless, cube-shaped storage room, the ceiling and 4 walls, including a door, are completely painted. The floor is not painted. If the painted area is equal to 80 square meters, what is the volume of the room, in cubic meters?

(A) 16
(B) 20
(C) 64
(D) 256
(E) 400

24. Two cylindrical candles of the same height and diameter burn at the same uniform rate. Each takes 4 hours to be consumed. If the first candle is lit at 8:00 p.m. and the second at 9:00 p.m., at what time will the second candle be exactly 3 times as tall as the first?

(A) 9:30 p.m.
(B) 10:00 p.m.
(C) 10:30 p.m.
(D) 11:00 p.m.
(E) 11:30 p.m.



Note: Figure not drawn to scale.

25. On line segment AE above, if the lengths of AC, BD, and CE are equal, which of the following must be true?

I. C is the midpoint of AE.
II. B is the midpoint of AC.
III. BC and DE are equal in length.

(A) I only
(B) III only
(C) I and II only
(D) I and III only
(E) I, II, and III

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

SECTION 4 Time—30 minutes 40 Questions

For each question in this section, choose the best answer and fill in the corresponding oval on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. TURBULENCE: (A) misconduct (B) calmness (C) sincerity (D) flexibility (E) prominence

2. MALICE: (A) familiarity (B) confidence (C) kindheartedness (D) good health (E) artistic talent

3. EXPANSIVE: (A) gracious (B) reserved (C) meticulous (D) unpopular (E) irresponsible

4. LACKADAISICAL: (A) unwholesome (B) unreliable (C) honest (D) energetic (E) pleasant

5. PALTRY: (A) frugal (B) impartial (C) superior (D) desperate (E) prudish

6. CORROBORATE: (A) challenge (B) adorn (C) display privately (D) restore completely (E) leave secretly

7. EXTEMPORANEOUS:
(A) obviously rehearsed
(B) correctly estimated
(C) rarely perceived
(D) accurately balanced
(E) extremely awkward

8. DIVEST: (A) ascend (B) confide (C) implicate (D) lapse (E) endow

9. COHERE: (A) splinter (B) absorb (C) detest (D) ignore (E) reproach

10. CELERITY: (A) slowness (B) foolishness (C) hardness (D) coldness (E) unhappiness

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that, when inserted in the sentence, best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. . respectable (B) extensive. . moderate (C) sophisticated. . amateur (D) risqué. . crude (E) perfect. . spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. A judgment made before all the facts are known must be called ----.

(A) harsh (B) deliberate (C) sensible (D) premature (E) fair

12. Because science is a discipline in which error is best discovered by peer review, it is appropriate that knowledgeable readers of scientific articles ---- what appear to be ---- interpretations of data.

(A) uphold. . untenable
(B) fabricate. . implausible
(C) ignore. . infallible
(D) question. . improbable
(E) praise. . equivocal

13. Although she knew that ---- was considered unfashionable in some circles, curator Estela Delgado was ---- to express her joy at the public's generous contributions to the two museums.

(A) exuberance. . unwilling
(B) restraint. . compelled
(C) spontaneity. . prepared
(D) decorum. . eager
(E) enthusiasm. . unashamed

GO ON TO THE NEXT PAGE

14. The author ---- the idea that someone would revive the incidents of his early life, and so he instructed his friends and associates to ---- any potential biographer.

(A) ridiculed. .respect (B) deplored. .forgive
(C) abhorred. .frustrate (D) savored. .avoid
(E) dismissed. .dissuade

15. The new instrument proved to be of little usefulness to the geologists; because it had no midrange sensitivity, it only registered disturbances that were either inconsequential or ----.

(A) cataclysmic (B) improbable
(C) subjective (D) instantaneous
(E) unintelligible

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN:BOREDOM :: (A) dream:sleep
(B) anger:madness (C) smile:amusement
(D) face:expression (E) impatience:rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. CYLINDRICAL : ROLLER ::
(A) mechanical : engine (B) spherical : globe
(C) protruding : knob (D) nuclear : reactor
(E) circular : path

17. PATENT : INVENTION :: (A) trigger : firearm
(B) key : lock (C) vault : money
(D) copyright : book (E) door : building

18. SWEEP : FLOOR :: (A) pave : street
(B) wipe : table (C) prune : shrub
(D) repair : car (E) mow : lawn

19. LARYNX : NECK :: (A) joint : elbow
(B) heart : chest (C) splint : bone
(D) cavity : tooth (E) indigestion : stomach

20. EXCOMMUNICATION : CHURCH ::
(A) expulsion : school (B) sermon : congregation
(C) execution : victim (D) dismissal : student
(E) extortion : business

21. EMIT : ODOR :: (A) reflect : shadow
(B) radiate : heat (C) incline : angle
(D) discharge : vent (E) measure : size

22. COMPOST : FERTILIZE :: (A) topsoil : erode
(B) scalpel : suture (C) anchor : hold
(D) detergent : stain (E) thermostat : install

23. SKULK : MOVE :: (A) blink : wink
(B) weep : grieve (C) peek : observe
(D) shake : quiver (E) cower : withdraw

24. STAGE : PLAY :: (A) rostrum : speech
(B) camera : film (C) orchestra : band
(D) theater : drama (E) hall : dance

25. OBDURATE : PERSUADE ::
(A) incorrigible : reform
(B) dour : displease
(C) crass : offend
(D) despondent : provoke
(E) gullible : convince

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Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

Uncle Harold claimed that he had been shot right between the eyes, in France, during the First World War. He told me so himself.

"Right between the eyes," he said. "See this scar?"

"I don't see any scar," I said.

"It's probably faded by now," he said. "It's been a long time ago."

It didn't matter that my mother called him "the biggest liar God ever sent down the pike." I found him irresistible. It was his refusal to spoil a good story by slavish adherence to fact that enchanted me.

He was not a tall man, but the Marines had taught him to carry himself erectly and to measure people with the grave, cool arrogance of authority. Though he now shoveled dirt for a living, he was always immaculately manicured by the time he sat down to supper. In this polished man of the world—suits pressed to razor sharpness, every hair in place—I began to detect a hidden boy, in spirit not unlike myself, though with a love for mischief which had been subdued in me by too much melancholy striving to satisfy my mother's notions of manhood.

I understood that Uncle Harold was not a liar but a teller of stories and a romantic. I no longer received his stories with total credulity, but listened for the pleasure of watching his imagination at play. We were two romancers whose desire for something more fanciful than the humdrum of our drab neighborhood was beyond the grasp of unimaginative people like my mother.

26. Which of the following titles is most suitable for the passage?

(A) Uncle Harold's Childhood and Adulthood
(B) A Polished Man of the World
(C) Hardships of a Struggling Family
(D) My Career as a Storyteller
(E) Not a Liar But a Romantic

27. The passage implies that as the boy learned the truth, his reaction to Uncle Harold's stories became one of

(A) delight in the flamboyance of his uncle's imagination
(B) respect for the danger in his uncle's earlier experiences
(C) disappointment in the deliberate deceptions practiced by his uncle
(D) regret for the persistent habits of his uncle
(E) resentment at the disturbing immaturity displayed by his uncle

28. The relationship between Uncle Harold and the author is best described as one of

(A) writer and reviewer
(B) director and actor
(C) employer and employee
(D) friendly rivals
(E) kindred spirits

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We are accustomed to thinking that great leaps forward in history occur because someone came up with a better way of doing things. Actually, these so-called better ways are in reality only different ways of doing things occasioned by the need to adjust to less easily exploitable sources of energy. Textbooks generally attribute the transition from the medieval era to the modern age to a great awakening of the human mind, and scholars debate the significance of the Protestant Reformation, the rise of the bourgeoisie, and the opening of trade routes in the great metamorphosis that occurred, but few admit that the fundamental cause of the Industrial Revolution was a switch from one energy base to another.

Today the sight of open countryside in Europe makes it difficult to imagine that in the fourth century a dense forest blanketed the continent from the Alps to the Carpathian Mountains. Wood in abundance provided the energy for the medieval way of life. A steady increase in the human population, however, between the ninth and twelfth centuries spurred the opening up of large tracts of land for cultivation, greatly reducing the available supply of wood. By the thirteenth century, wood had become so scarce that a search for alternative sources of energy began, eventually leading to coal.

Since it is more difficult to mine and process coal than it is to cut down trees, the switch from wood to coal initiated a long series of technological innovations. The process of mining coal, for example, was facilitated by the introduction of the steam engine. Such innovations laid the foundation for the industrial era that emerged.

29. The primary purpose of the passage is to

(A) support a theory
(B) criticize a method
(C) defend a way of life
(D) refute an opponent
(E) predict a trend

30. The passage suggests that the Industrial Revolution might not have happened as it did were it not for the

(A) competitiveness among technologically minded inventors
(B) emergence of commercialism and the middle class
(C) inefficient management of resources before the fourth century
(D) exorbitant price of high-quality wood in the thirteenth century
(E) steady rise in population from the ninth to the twelfth century

31. Which of the following maxims is most compatible with the view of change that is presented in the passage?

(A) There is nothing new under the sun.
(B) All is for the best in the best of all possible worlds.
(C) Little strokes fell great oaks.
(D) Necessity is the mother of invention.
(E) Human history is a history of ideas.

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About 10,000 years ago, as glaciers retreated into Canada, North America lost most of its large animals. The casualty list included mammoths, many species of horses and camels, the giant beaver, and others, totaling over 100 species. The question of what caused this late Pleistocene extinction has provoked a storm of controversy.

One widely held hypothesis is that sudden climatic change was responsible for the abrupt extinctions. My own hypothesis is that prehistoric human hunters were responsible. This view is neither new nor widely held. To discount the hypothesis one need simply identify a major wave of extinctions anywhere in the world in the late Pleistocene age prior to the hunters' arrival. To date, such evidence has not been found. In fact, the chronological sequence of extinction closely follows human footsteps, occurring first in Africa and southern Asia, next in Australia, then through northern Eurasia and into North and South America, much later in the West Indies, and finally, during the last 1,000 years, in Madagascar and New Zealand. The pattern shows that late Pleistocene extinction did not occur in all locations at the same time, as it would have if there had been a sudden climatic change or perhaps a cataclysmic destruction of the Earth's atmosphere by lethal radiation from cosmic-ray bombardment, another common hypothesis. Since no synchronous destruction of plants or of plant communities is known, the long-held belief that climatic change caused the extinction lacks credibility.

32. The central argument of the passage is that

(A) all the evidence points to the arrival of humans in North America about 10,000 years ago
(B) large animals became extinct in North America 10,000 years ago
(C) sudden climatic change does not affect the survival of a species
(D) no species became extinct before the arrival of human hunters
(E) human hunters caused the extinction of large animals in the late Pleistocene age

33. According to the passage, which of the following is (are) true of the author's hypothesis?

I. The author is the first person to propose it.
II. No evidence has yet been found to disprove it conclusively.
III. Few scholars interested in the subject have endorsed it.

(A) I only (B) I and II only
(C) I and III only (D) II and III only
(E) I, II, and III

34. According to the passage, which of the following would invalidate the author's argument?

(A) The discovery of major Pleistocene extinctions that occurred before the arrival of humans
(B) Evidence that a bombardment of cosmic rays had occurred after humans had reached North America
(C) Proof that North America has experienced little climatic change
(D) Discovery of a species of large animal that survived the arrival of humans in North America
(E) Proof that humans had come to North America from Europe

35. All of the following are central to the author's argument EXCEPT:

(A) The number of species of large animals native to North America decreased.
(B) Cosmic-ray bombardment would have affected every place on Earth at the same time.
(C) Many animals became extinct before the late Pleistocene age.
(D) Prehistoric humans hunted large animals.
(E) A dramatic and widespread climatic change would destroy both plant and animal species.

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An architect's stupidity is more dangerous than any other, for its inescapable influence crushes the sensitive individual by the disorder of proportions. It is not a matter of good taste or bad; it is a matter of the site that exhausts, that envenoms, that debases, that casts its spells and curses in silence, in secret. A hotel, a ship can cause strange ravages. You cannot trace the source of the discomfort that dissipates your resources, but your soul slowly stiffens and loses its adaptability. The malaise is impossible to analyze. At first you laugh at ugliness; it intrigues you, revolts you. Gradually it poisons you; your organism refuses to prosper; it squints, it limps, it dies.

In most cases the picturesque and the fantastic overcome us with fatigue and boredom. Forms, lines, and colors exert a power that the Occidental controls but imperfectly, and that the Oriental uses to achieve specific and predetermined effects. An Indian temple, a Chinese pagoda can hypnotize, bewitch, excite, lull, all by the use of volumes, curves, perspectives.

A monument that *serves*, or that has served, never fatigues us. The Colosseum served, the Acropolis served, the Sphinx served, ... That is why they delight us. It is not necessary to know what they served *for*, or to take advantage of their services. The fact that they were born of a need, that a purpose directed those who built them and obliged them to submit to certain rules, clears these monuments of all disorder, all frivolity. Whether they were intended to astound, to thank, to overpower, to assure the survival of the dead by the resemblance of a double, or, by that resemblance, to terrify tomb robbers, the point of departure is not a matter of chance. The great epochs never confront us with works of aesthetes. A scarecrow frightens birds, not us, but the very necessity of achieving a successful result inspires the farmer who devises it, dispensing him from being decorative. That is the beauty of the scarecrow. And of African masks, totems, the Sphinxes of Egypt.

Powerful and almost always secret motives are to be found at the origin of a thousand details that weave the seething beauty of nature. A singularity may appear gratuitous to us, but its expressive force always conceals roots. The same holds for the Orient, its architecture, its art; once we know the source of its nuances we no longer find them odd, quaint. The tourists who travel there must realize that they are passing through a ritual of which they understand nothing. Let this realization make them wary of judging, and strip them once and for all of our Western impertinence. Let nothing that disconcerts them make them smile; let them respect signs that lose all their decorative naïveté as soon as our minds divine their meaning.

36. It can be inferred from the passage that an ill-conceived building is especially offensive because of the

(A) insidiousness with which it impinges on one's consciousness
(B) undeserved profit its owner is likely to realize
(C) irrevocable harm it can do to the public's perception of architecture
(D) demolition of better buildings to make way for its construction
(E) impossibility of improving even its least problematic features

37. A stylistic device by which the author emphasizes the effects of oppressive architecture on a sensitive individual is the

(A) accumulation of short verb phrases
(B) juxtaposition of contradictory statements
(C) alternation between points of view
(D) response to implied rhetorical questions
(E) repetition of adjectives with negative connotations

38. The author argues that the monuments that delight us are those that

(A) were created to endure for thousands of years
(B) are similar to those our own era might create
(C) were built to serve a particular social purpose
(D) are inherently interesting by virtue of their proportions
(E) were intended to mystify all who encountered them

39. It can be inferred from the final paragraph that the author believes people who consider unfamiliar art or architecture merely curious or quaint to have

(A) no hope of enjoying foreign travel
(B) a seriously uninformed attitude
(C) no potential for any spiritual growth
(D) a need to be entertained rather than impressed
(E) no understanding of human nature

40. The passage as a whole combines

(A) an analysis of history with an evaluation of creativity
(B) a personal reminiscence with general observations
(C) a description of monuments with prescriptions for travel
(D) a theory of architecture with an appreciation of the East
(E) a critique of the West with a nostalgic evocation of the past

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

SECTION 5 Time—30 minutes 35 Questions

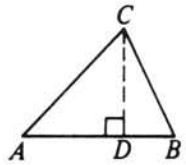
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and fill in the corresponding oval on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definition of symbols:

$=$ is equal to	$\leqq$ is less than or equal to
$\neq$ is unequal to	$\geqq$ is greater than or equal to
$<$ is less than	$\parallel$ is parallel to
$>$ is greater than	$\perp$ is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures that accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

1. If $r = 3$, then $(2r)^2 =$

(A) 10
(B) 12
(C) 18
(D) 25
(E) 36

2. If $x + y = 10$ and $x - y = 2$, then $xy =$

(A) 24
(B) 20
(C) 16
(D) 12
(E) 8

3. Brand X granola bars are packed 12 to a box, and each granola bar supplies 2 grams of protein. How many such boxes of granola bars would supply 72 grams of protein?

(A) 3
(B) 10
(C) 12
(D) 24
(E) 36

4. $9 + \frac{9}{100} + \frac{9}{10{,}000} =$

(A) 9.99
(B) 9.909
(C) 9.099
(D) 9.0909
(E) 9.0099

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(6)	4
9	7
(8)	5
(4)	13
8	6

5. In the figure above, the average (arithmetic mean) of the numbers in each column is k. If the three circled numbers are moved from the left to the right column, which of the following combinations of numbers can then be moved from the right to the left column so that k remains the average of the numbers in each column?

(A) 6, 13
(B) 4, 5, 6
(C) 4, 7, 5
(D) 4, 7, 6
(E) 7, 5, 6



6. The figure above shows an accurate clock. If Eric leaves his office exactly 3 hours and 20 minutes after the time shown, which of the following will be the measure of the angle that the minute hand forms with the hour hand when Eric leaves his office?

(A) 60°
(B) 90°
(C) 120°
(D) 150°
(E) 180°

7. The ratio of the length of a rectangular floor to its width is 3:2. If the length of the floor is 12 meters, what is the perimeter of the floor, in meters?

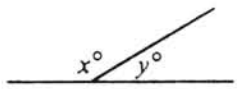
(A) 20
(B) 40
(C) 60
(D) 96
(E) 120

GO ON TO THE NEXT PAGE

Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet fill in oval

A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

AN E RESPONSE WILL NOT BE SCORED.

EXAMPLES	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ Ⓔ
E2.	 $180 - x$	y	Ⓐ Ⓑ ● Ⓓ Ⓔ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ● Ⓔ

Notes:

1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. In a given question, a symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

	Column A	Column B
8.	$(-2)(-8)$	$(-2) + (-8)$
9.	$(3 \times 10^5) + (4 \times 10^3)$	$(3 \times 10^5) + (5 \times 10^2)$
10.	The number of days in 8 weeks	The number of hours in $2\frac{1}{2}$ days

$a < 0$

	Column A	Column B
11.	$\frac{1}{6}$	$\frac{1}{a^2}$

CARS SOLD BY DEALER X	
1983	
1982	

In 1983 Dealer X sold 270 more cars than in 1982.

	Column A	Column B
12.	The number of cars each  represents	100
13.	$\frac{1}{2} \times \frac{1}{2}$	$\frac{\frac{1}{2}}{2}$

$$\frac{x}{12} = \frac{5}{11}$$

	Column A	Column B
14.	x	5

	Column A	Column B
15.	$\begin{array}{r} 23 \\ 16 \\ 57 \\ +\ x \\ \hline 108 \end{array}$	$\begin{array}{r} 23 \\ 16 \\ 27 \\ +\ y \\ \hline 108 \end{array}$
	x	y

P and Q are two points on the circumference of a circle with center O.

	Column A	Column B
16.	Length of PO	Length of QO

$$a^3 = b^3$$

	Column A	Column B
17.	ab	a^2



The tick marks on the number line are equally spaced.

	Column A	Column B
18.	The coordinate of point P	0.4256
19.	$2(x - 4y)$	$2x - 4y$

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5

SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS

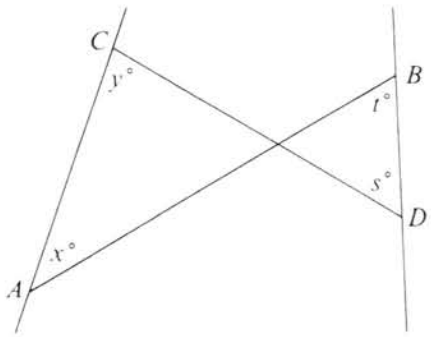
Answer: A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

AN E RESPONSE WILL NOT BE SCORED.

Column A | Column B

$s = 6q$, where q is a positive integer, and
$s = 2r$, where r is a positive integer.

20. s | 12

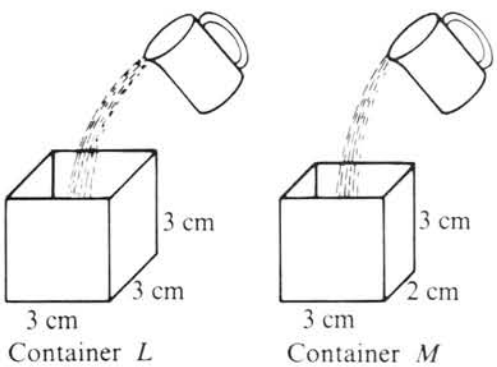


Line segments AB and CD intersect as shown.

21. $x + y$ | $s + t$

$x - y > 0$
$x + y < 0$

22. y | 0

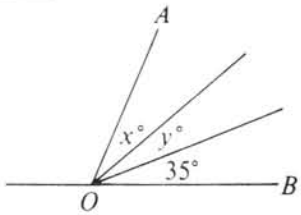


X cubic centimeters of water was poured into each of the empty rectangular containers above. The containers rest on a level surface.

$0 < X < 18$

23. The height of the water in container L | The height of the water in container M

Column A | Column B



Note: Figure not drawn to scale.
$\angle AOB = 80°$ and $y > 10$

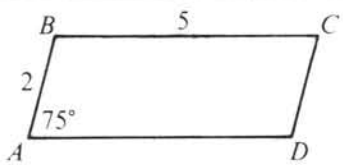
24. x | 35

$x > 0$

25. $3 + \dfrac{1}{x + \frac{1}{2} + \frac{1}{4}}$ | 4

---

26. The greatest possible number of points common to a triangle and a circle | 3



27. The area of parallelogram $ABCD$ | 10

GO ON TO THE NEXT PAGE

Solve each of the remaining problems in this section using any available space for scratchwork. Then decide which is the best of the choices given and fill in the corresponding oval on the answer sheet.

28. A triangle has sides with lengths $x - 2$, $2x - 3$, and $x + 1$. If the perimeter of the triangle is 20, what is the length of its longest side?

(A) 9
(B) 8
(C) 7
(D) 6
(E) 5

Questions 29-30 refer to the following definition.

S is the set of all integers that can be written as $n^2 + 1$ where n is a nonzero integer.

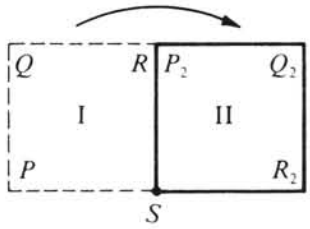
29. Which of the following integers is in S?

(A) 16
(B) 28
(C) 35
(D) 39
(E) 50

30. If x and y are two different integers in S, which of the following must be the square of an integer?

I. $(x - 1)(y - 1)$
II. $4(x - 1)$
III. $x^2 + y^2$

(A) None
(B) I only
(C) II only
(D) III only
(E) I and II



31. In the figure above, square $PQRS$, initially in position I, has been rotated clockwise about point S to position II. If P_2, Q_2, and R_2 are the second positions of P, Q, and R, respectively, and if a side of the square is 1, what is the length of the path followed by P in rotating to P_2?

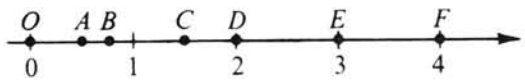
(A) $\frac{\pi}{4}$
(B) 1
(C) $\frac{\pi}{2}$
(D) 2
(E) π

32. If x and y are positive integers, x divided by 7 leaves a remainder of 3, and y divided by 7 leaves a remainder of 4, what is the remainder when xy is divided by 7?

(A) 0
(B) 1
(C) 2
(D) 5
(E) 6

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5



33. On the number line above, the ratio of distance OB to distance OA is closest to which of the following distance ratios?

(A) $\frac{OC}{OB}$

(B) $\frac{OD}{OB}$

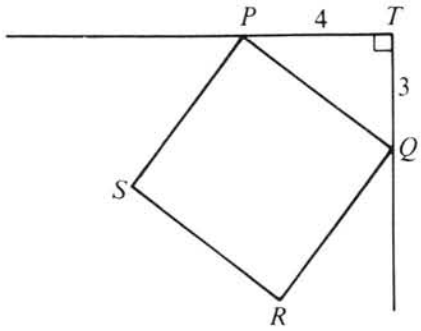
(C) $\frac{OE}{OD}$

(D) $\frac{OE}{OC}$

(E) $\frac{OF}{OD}$

34. The average (arithmetic mean) weight of 3 boxes is $75\frac{1}{3}$ pounds. If each box weighs at least 74 pounds, what is the greatest possible weight, in pounds, of one of these boxes?

(A) 75
(B) 76
(C) 77
(D) 78
(E) 79



35. In the figure above, if $PQRS$ is a square, $PT \perp QT$, $QT = 3$, and $PT = 4$, then $RT =$

(A) $5\sqrt{2}$

(B) $2\sqrt{15}$

(C) $\sqrt{58}$

(D) $\sqrt{65}$

(E) $\sqrt{66}$

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

Correct Answers for Scholastic Aptitude Test Form Code 6K			
VERBAL		MATHEMATICAL	
Section 1	Section 4	Section 2	Section 5
1. D	1. B	1. A	1. E
2. A	2. C	2. E	2. A
3. C	3. B	3. D	3. A
4. B	4. D	4. B	4. D
5. A	5. C	5. A	5. E
6. A	6. A	6. E	6. E
7. E	7. A	7. D	7. B
8. D	8. E	8. B	*8. A
9. D	9. A	9. C	*9. A
10. C	10. A	10. D	*10. B
11. D	11. D	11. C	*11. D
12. A	12. D	12. A	*12. A
13. E	13. E	13. E	*13. C
14. E	14. C	14. B	*14. A
15. E	15. A	15. B	*15. B
16. B	16. B	16. D	*16. C
17. C	17. D	17. D	*17. C
18. D	18. B	18. C	*18. A
19. E	19. B	19. D	*19. D
20. D	20. A	20. A	*20. D
21. E	21. B	21. D	*21. C
22. D	22. C	22. C	*22. B
23. C	23. C	23. C	*23. B
24. A	24. A	24. E	*24. B
25. A	25. A	25. D	*25. D
26. B	26. E		*26. A
27. E	27. A		*27. B
28. B	28. E		28. A
29. C	29. A		29. E
30. A	30. E		30. E
31. A	31. D		31. C
32. C	32. E		32. D
33. B	33. D		33. C
34. A	34. A		34. D
35. E	35. C		35. C
36. E	36. A		
37. B	37. A		
38. E	38. C		
39. C	39. B		
40. D	40. D		
41. C			
42. E			
43. D			
44. E			
45. A			

*Indicates four-choice questions. (All of the other questions are five-choice.)

The Scoring Process

Machine-scoring is done in three steps:

- *Scanning.* Your answer sheet is "read" by a scanning machine and the oval you filled in for each question is recorded on a computer tape.
- *Scoring.* The computer compares the oval filled in for each question with the correct response. Each correct answer receives one point; omitted questions do not count toward your score. For each wrong answer, a fraction of a point is subtracted to correct for random guessing. For questions with five answer choices, one-fourth of a point is subtracted for each wrong response; for questions with four answer choices, one-third of a point is subtracted for each wrong response. The SAT-verbal test has 85 questions with five answer choices each. If, for example, a student has 44 right, 32 wrong, and 9 omitted, the resulting raw score is determined as follows:

$$44 \text{ right} - \frac{32 \text{ wrong}}{4} = 44 - 8 = 36 \text{ raw score points}$$

Obtaining raw scores frequently involves the rounding of fractional numbers to the nearest whole number. For example, a raw score of 36.25 is rounded to 36, the nearest whole number. A raw score of 36.50 is rounded upward to 37.

- *Converting to reported scaled score.* Raw test scores are then placed on the College Board scale of 200 to 800 through a process that adjusts scores to account for minor differences in difficulty among different editions of the test. This process, known as equating, is performed so that a student's reported score is not affected by the edition of the test taken nor by the abilities of the group with whom the student takes the test. As a result of placing SAT scores on the College Board scale, scores earned by students at different times can be compared. For example, an SAT-verbal score of 400 on a test taken at one administration indicates the same level of developed verbal ability as a 400 score obtained on a different edition of the test taken at another time.

How to Score the Test

SAT-Verbal Sections 1 and 4

Step A: Count the number of correct answers for *section 1* and record the number in the space provided on the worksheet on the next page. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for *section 4* and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: To obtain C, add subtotal A to subtotal B, keeping any decimals. Enter the resulting figure on the worksheet.

Step D: To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step E: To find your reported SAT-verbal score, look up the total raw verbal score you obtained in step D in the conversion table on page 196. Enter this figure on the worksheet.

SAT-Mathematical Sections 2 and 5

Step A: Count the number of correct answers and the number of incorrect answers for *section 2* and record the numbers in the spaces provided on the worksheet. To determine the subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal C}$$

Step D: To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals. Enter the resulting figure on the worksheet.

Step E: To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step F: To find your reported SAT-mathematical score, look up the total raw mathematical score you obtained in E in the conversion table on page 196. Enter this figure on the worksheet.

SAT SCORING WORKSHEET

FORM CODE 6K

SAT-Verbal Sections

A. Section 1: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal A

B. Section 4: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw score (Rounded to nearest whole number) _____ D

E. SAT-verbal reported scaled score (See the conversion table on page 196.) [ ] SAT-verbal score

SAT-Mathematical Sections

A. Section 2: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal A

B. Section 5: Questions 1 through 7 and 28 through 35 (5-choice) _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal B

C. Section 5: Questions 8 through 27 (4-choice) _____ (no. correct) − ⅓ (_____) (no. incorrect) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw score (Rounded to nearest whole number) _____ E

F. SAT-mathematical reported scaled score (See the conversion table on page 196.) [ ] SAT-math score

Score Conversion Table
Scholastic Aptitude Test
Form Code 6K

Raw Score	College Board Reported Score		Raw Score	College Board Reported Score	
	SAT-Verbal	SAT-Math		SAT-Verbal	SAT-Math
85	800		40	460	590
84	780		39	450	580
83	760		38	440	570
82	750		37	440	560
81	740		36	430	550
80	730		35	430	540
79	720		34	420	530
78	710		33	410	520
77	700		32	410	510
76	700		31	400	500
75	690		30	390	490
74	680		29	390	480
73	670		28	380	480
72	660		27	380	470
71	650		26	370	460
70	650		25	360	450
69	640		24	360	440
68	630		23	350	430
67	620		22	340	420
66	620		21	340	410
65	610		20	330	410
64	600		19	320	400
63	600		18	320	390
62	590		17	310	380
61	580		16	300	370
60	580	800	15	300	360
59	570	780	14	290	360
58	560	770	13	280	350
57	560	760	12	280	340
56	550	750	11	270	330
55	540	740	10	260	320
54	540	730	9	250	320
53	530	720	8	250	310
52	530	710	7	240	300
51	520	690	6	230	290
50	510	690	5	230	280
49	510	680	4	220	280
48	500	670	3	210	270
47	500	660	2	200	260
46	490	650	1	200	250
45	480	640	0	200	250
44	480	630	−1	200	240
43	470	620	−2	200	230
42	470	610	−3	200	220
41	460	600	−4	200	210
			−5	200	210
			−6	200	200
			or below		